In[1]:=
```
(*1(a)*)
s = 1;
Do[s = s * k, {k, 1, 12}];
s
```

Out[3]= 479 001 600

In[4]:=
```
s = 1; i = 1;
While[i ≤ 12, s = s * i; i = i + 1]
s
```

Out[6]= 479 001 600

In[7]:=
```
s = 1;
For[k = 1, k ≤ 12, k++, s = s * k]
s
```

Out[9]= 479 001 600

In[10]:=
```
(*1(b)(i)*)
LogicalExpand[¬ (p ⋀ q)] == LogicalExpand[¬ p ⋁ ¬ q]
```

Out[10]= True

In[11]:=
```
(*1(b)(ii)*)
LogicalExpand[¬ (p ⋁ q)] == LogicalExpand[¬ p ⋀ ¬ q]
```

Out[11]= True

In[12]:=
```
(*2(a)*)
s = Select[Range[500], Mod[#, 9] ≠ 0 &];
Partition[s, 5, 5, 1, {}] // TableForm
```

Out[13]//TableForm=

1	2	3	4	5
6	7	8	10	11
12	13	14	15	16
17	19	20	21	22
23	24	25	26	28
29	30	31	32	33
34	35	37	38	39
40	41	42	43	44
46	47	48	49	50
51	52	53	55	56
57	58	59	60	61
62	64	65	66	67
68	69	70	71	73
74	75	76	77	78

```
79  80  82  83  84
85  86  87  88  89
91  92  93  94  95
96  97  98  100 101
102 103 104 105 106
107 109 110 111 112
113 114 115 116 118
119 120 121 122 123
124 125 127 128 129
130 131 132 133 134
136 137 138 139 140
141 142 143 145 146
147 148 149 150 151
152 154 155 156 157
158 159 160 161 163
164 165 166 167 168
169 170 172 173 174
175 176 177 178 179
181 182 183 184 185
186 187 188 190 191
192 193 194 195 196
197 199 200 201 202
203 204 205 206 208
209 210 211 212 213
214 215 217 218 219
220 221 222 223 224
226 227 228 229 230
231 232 233 235 236
237 238 239 240 241
242 244 245 246 247
248 249 250 251 253
254 255 256 257 258
259 260 262 263 264
265 266 267 268 269
271 272 273 274 275
276 277 278 280 281
282 283 284 285 286
287 289 290 291 292
293 294 295 296 298
299 300 301 302 303
304 305 307 308 309
310 311 312 313 314
316 317 318 319 320
321 322 323 325 326
327 328 329 330 331
```

```
332 334 335 336 337
338 339 340 341 343
344 345 346 347 348
349 350 352 353 354
355 356 357 358 359
361 362 363 364 365
366 367 368 370 371
372 373 374 375 376
377 379 380 381 382
383 384 385 386 388
389 390 391 392 393
394 395 397 398 399
400 401 402 403 404
406 407 408 409 410
411 412 413 415 416
417 418 419 420 421
422 424 425 426 427
428 429 430 431 433
434 435 436 437 438
439 440 442 443 444
445 446 447 448 449
451 452 453 454 455
456 457 458 460 461
462 463 464 465 466
467 469 470 471 472
473 474 475 476 478
479 480 481 482 483
484 485 487 488 489
490 491 492 493 494
496 497 498 499 500
```

In[14]:=
```
(*2(b)*)
Clear[f, x, h]
f[x_] := 4 x^3 + 3 x^2 - x + 2
a = Collect[f[x + h], x];
s = Solve[Coefficient[a, x^2] == 0, h];
(a /. s[[1]]) == 0 // Expand
```

Out[18]= $\frac{19}{8} - \frac{7\,x}{4} + 4\,x^3 == 0$

In[19]:=
```
(*3(a)*)
p = (x - 1) (x - 2)^2 (x - 3)^3;
q = (x - 1)^2 (x - 2) (x - 3)^4;
a = Factor[PolynomialGCD[p, q]]
```

Out[21]= $(-3 + x)^3\,(-2 + x)\,(-1 + x)$

In[22]:=
```
b = Factor[PolynomialLCM[p, q]]
```

Out[22]= $(-3 + x)^4\,(-2 + x)^2\,(-1 + x)^2$

In[23]:=
```
Expand[a * b] == Expand[p * q]
```

Out[23]= True

In[24]:=
```
(*3(b)*)
s = {}; k = 1;
While[Prime[k] ≤ 600, s = Append[s, Prime[k]]; k++]
s1 = {}; k = 1;
While[Fibonacci[k] ≤ 600, s1 = Append[s1, Fibonacci[k]]; k++]
s2 = Intersection[s, s1]
```

Out[28]= {2, 3, 5, 13, 89, 233}

In[29]:=
```
(*4(i)*)
Clear[A]
A = {{1, 2, 2}, {1, 2, -1}, {-1, 1, 4}};
b = CharacteristicPolynomial[A, x];
c = CoefficientList[b, x];
d = c[[1]] * IdentityMatrix[3] + c[[2]] * A +
   c[[3]] * MatrixPower[A, 2] + c[[4]] * MatrixPower[A, 3];
If[d == {{0, 0, 0}, {0, 0, 0}, {0, 0, 0}}, Print["Cayley-Hamilton theorem is verified"],
 Print["Not Verified"]]
```

Cayley-Hamilton theorem is verified

In[35]:=
```
(*4(ii)*)
1 / c[[1]] * (c[[2]] * IdentityMatrix[3] - c[[3]] * A + c[[4]] * MatrixPower[A, 2]) // MatrixForm
```

Out[35]//MatrixForm=

$$\begin{pmatrix} -\frac{23}{9} & -\frac{22}{9} & -\frac{22}{9} \\ -\frac{11}{9} & -\frac{34}{9} & \frac{11}{9} \\ \frac{11}{9} & -\frac{11}{9} & -\frac{56}{9} \end{pmatrix}$$

In[36]:=
```
(*4(iii)*)
Print["Eigenvalues:", Eigenvalues[A]]
Print["Eigenvectors:", Eigenvectors[A]]
```

Eigenvalues:{3, 3, 1}

Eigenvectors:{{1, 0, 1}, {1, 1, 0}, {2, -1, 1}}

In[38]:=
```
(*4(iv)*)
RRank = MatrixRank[A];
CRank = MatrixRank[Transpose[A]];
If[RRank == CRank, Print["The Row rank and Column rank of A are equal."]]
```

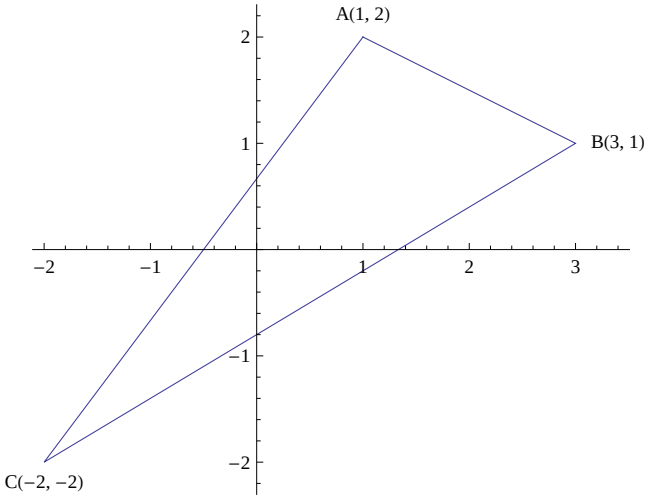
The Row rank and Column rank of A are equal.

In[41]:=
```
(*5(a)*)
A = {{2, 1, 1}, {1, -1, 2}, {1, 1, -1}}; Ax = {{8, 1, 1}, {1, -1, 2}, {4, 1, -1}}; Ay = {{2, 8, 1}, {1, 1, 2}, {1, 4, -1}}; Az = {{2, 1, 8}, {1, -1, 1}, {1, 1, 4}};
x = Det[Ax] / Det[A]; y = Det[Ay] / Det[A]; z = Det[Az] / Det[A];
Print["The required solution is (", x, ",", y, ",", z, ")"]
```

The required solution is (2,3,1)

In[44]:=
```
(*5(b)*)
pts = {{1, 2}, {3, 1}, {-2, -2}, {1, 2}};
pta = Graphics[Text["A(1, 2)", {1, 2.2}]];
ptb = Graphics[Text["B(3, 1)", {3.4, 1}]];
ptc = Graphics[Text["C(-2, -2)", {-2, -2.2}]];
g1 = ListPlot[pts, Joined → True];
Show[{pta, ptb, ptc, g1}, AxesOrigin → {0, 0}, Axes → True, AspectRatio → Automatic]
```

Out[49]=



In[50]:=
```
(*6(a)*)
p := 6 x^2 - 4 y^2 + 5 x * y + 7 x + 13 y - 3;
a = 6; b = -4; c = -3; h = 5 / 2; g = 7 / 2; f = 13 / 2;
d = a b c + 2 f g h - a f^2 - b g^2 - c h^2;
If[d == 0, Print["Represents a pair of straight line"],
 Print["Does not represent a pair of straight line"]]
```

Represents a pair of straight line

In[54]:=
```
Print["The point of intersection is (", (h * f - b * g)/(a * b - h^2), ",", (g * h - a * f)/(a * b - h^2), ")"]
```

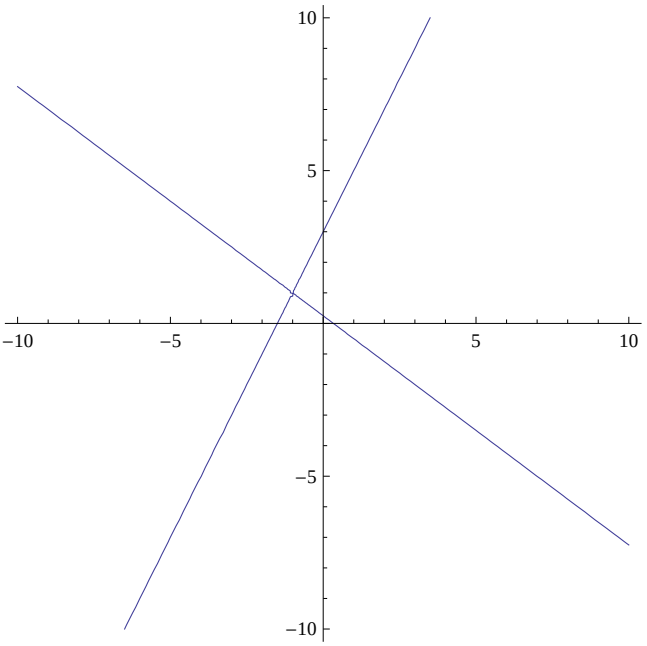
The point of intersection is (-1,1)

In[55]:=
```
h1 = a * b - h^2;
c1 = a + b;
If[c1 == 0, Print["The angle between the line is θ = ", π/2],
 Print["The angle between the line is θ = ", ArcTan[2 Sqrt[-h1]/c1] // N]]
```

The angle between the line is θ = 1.39094

In[58]:=
```
ContourPlot[p == 0, {x, -10, 10}, {y, -10, 10}, Axes → True, Frame → False]
```

Out[58]=



In[59]:=
```
(*6(b)(i)*)
u = {u1, u2, u3}; v = {v1, v2, v3}; w = {w1, w2, w3};
LHS = Cross[Cross[u, v], w] // Expand;
RHS = v (u.w) - u (v.w) // Expand;
LHS == RHS
```

Out[62]= True

In[63]:=
```
(*6(b)(ii)*)
a = {1, 2, -3};
Print["Angle with x-axis is ", VectorAngle[a, {1, 0, 0}]]
```

Angle with x-axis is $\text{ArcCos}\left[\frac{1}{\sqrt{14}}\right]$

In[65]:=
```
Print["Angle with y-axis is ", VectorAngle[a, {0, 1, 0}]]
```

Angle with y-axis is $\text{ArcCos}\left[\sqrt{\frac{2}{7}}\right]$

In[66]:=
```
Print["Angle with z-axis is ", VectorAngle[a, {0, 0, 1}]]
```

Angle with z-axis is $\text{ArcCos}\left[-\frac{3}{\sqrt{14}}\right]$

In[72]:=
```
(*7(a)*)
Clear[x, y]
y[x_] := x^2 + x - 2
slope = D[y[x], x]
```

Out[74]= $1 + 2\,x$

In[75]:=
```
m = slope /. {x → 2, y → 4}
```

Out[75]= 5

In[76]:=
```
x1 = 2; y1 = 4;
Print["Equation of tangent is ", y - y1 == m (x - x1) // Simplify]
```

Equation of tangent is $5\,x == 6 + y$

In[78]:=
```
Print["Equation of normal is ", y - y1 == -(1 / m) (x - x1) // Simplify]
```

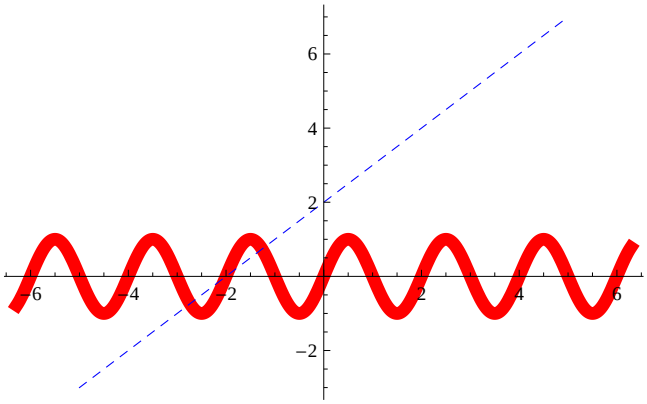
Equation of normal is $x + 5\,y == 22$

In[79]:=
```
(*7(b)*)
x1 = 1; y1 = -2; z1 = 3; x2 = 2; y2 = -3; z2 = 5;
a1 = 2; b1 = 3; c1 = 4; a2 = 3; b2 = 4; c2 = 5;
p = b1 * c2 - b2 * c1; m = c1 * a2 - c2 * a1; n = a1 * b2 - a2 * b1;
sd = Abs[p / Sqrt[p^2 + m^2 + n^2] * (x2 - x1) + m / Sqrt[p^2 + m^2 + n^2] * (y2 - y1) + n / Sqrt[p^2 + m^2 + n^2] * (z2 - z1)];
Print["The short distance is ", sd]
```

The short distance is $2\sqrt{\frac{2}{3}} + \frac{1}{\sqrt{6}}$

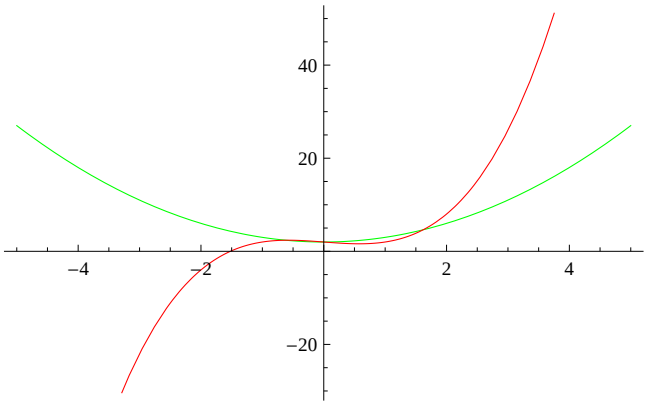
In[89]:=
```
(*8(a)*)
Clear[f, g]
f[x_] := Sin[π * x]
g[x_] := x + 2
p1 = Plot[f[x], {x, -2 π, 2 π}, PlotRange → All, PlotStyle -> {Thickness[0.02], Red}];
p2 = Plot[g[x], {x, -5, 5}, PlotRange → All, PlotStyle -> {Blue, Dashed}];
Show[{p1, p2}]
```

Out[94]=



In[95]:=
```
(*8(b)*)
f[x_] := x^2 + 2
g[x_] := x^3 - x + 2
Plot[{f[x], g[x]}, {x, -5, 5}, PlotStyle → {RGBColor[0, 1, 0], RGBColor[1, 0, 0]}]
```

Out[97]=



In[98]:=
```
(*9(a)*)
Clear[f, x]
f[x_] := Abs[x + 2] + Abs[x]
LHL = Limit[f[x], x → -2, Direction → 1];
RHL = Limit[f[x], x → -2, Direction → -1];
If[f[-2] == LHL == RHL, Print["f(x) is continuous at x=-2"],
 Print["f(x) is not continuous at x=-2"]]
```

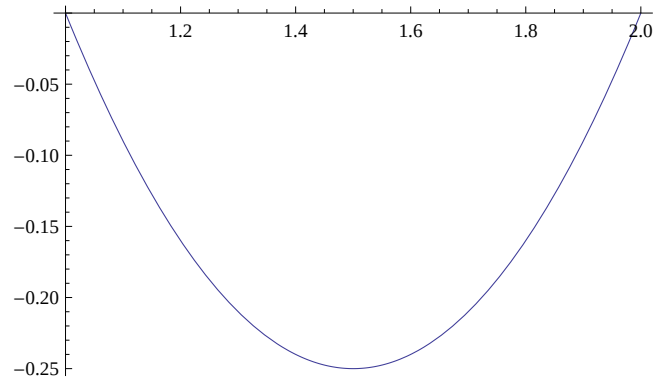
```
f(x) is continuous at x=-2
```

In[103]:=
```
(*9(b)*)
Clear[f, c]
f[x_] := x^2 - 3 x + 2
a = 1; b = 2;
f[a] == f[b] == 0
```

Out[106]=
```
True
```

In[107]:=
```
Plot[f[x], {x, a, b}]
```

Out[107]=



In[108]:=
```
s = FindRoot[f'[c] == 0, {c, 1}]
```

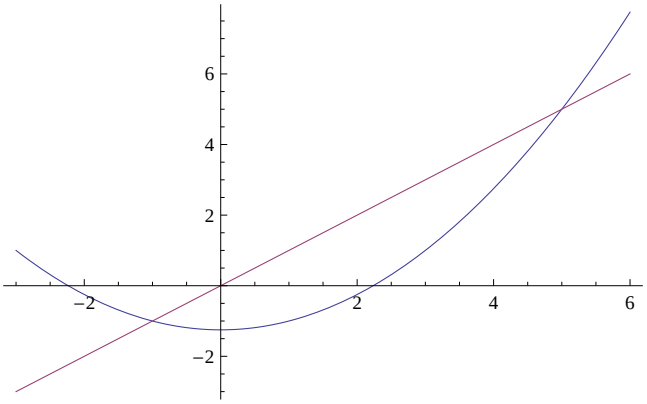
Out[108]=
```
{c → 1.5}
```

In[109]:=
```
If[a < c < b /. s, Print["Roll's theorem is verified."]]
```

```
Roll's theorem is verified.
```

In[110]:=
```
(*10(a)*)
Clear[y]
f1[y_] := 1/4 (y^2 - 5)
f2[y_] := y
Plot[{f1[y], f2[y]}, {y, -3, 6}]
```

Out[113]=



In[114]:=
```
Solve[f1[y] == f2[y]]
```

Out[114]=
```
{{y → -1}, {y → 5}}
```

In[115]:=
```
a = -1; b = 5;
Area = ∫_a^b (f2[y] - f1[y]) dy
```

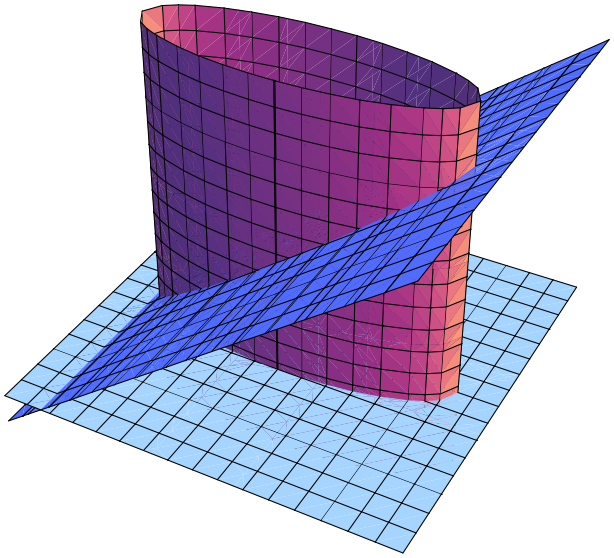
Out[116]=
```
9
```

In[117]:=
```
(*10(b)(i)*)
Clear[x, y, z]
∫_-4^4 ∫_-3^3 ∫_0^1 (x^2 + y^2 - z^2) dx dy dz
```

Out[118]=
```
-96
```

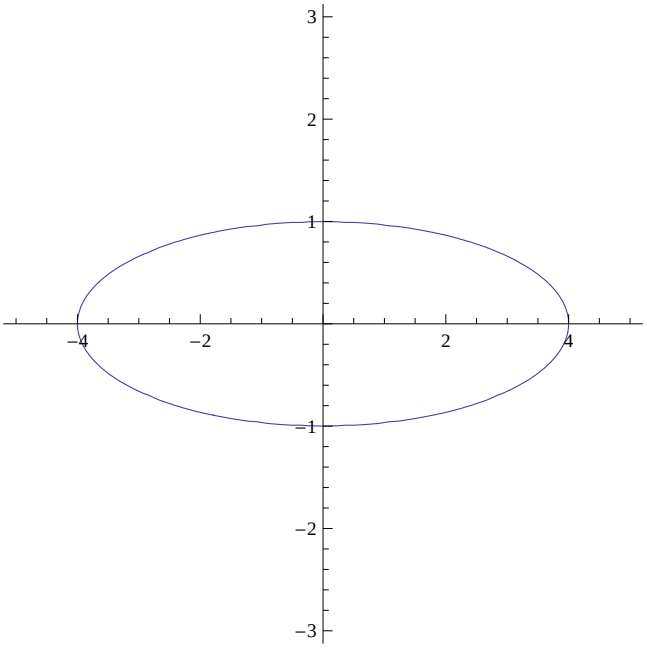
In[119]:=
```
(*10(b)(ii)*)
ContourPlot3D[{x^2 + 16 y^2 == 16, z == 0, z == x + 4},
 {x, -5, 5}, {y, -4, 4}, {z, -3, 10}, Boxed → False, Axes → False]
```

Out[119]=



In[120]:=
```
ContourPlot[x^2 + 16 y^2 == 16, {x, -5, 5}, {y, -3, 3}, Axes → True, Frame → False]
```

Out[120]=



In[121]:= **Solve[x² + 16 y² == 16, x]**

Out[121]= $\left\{\left\{x \to -4\sqrt{1-y^2}\right\}, \left\{x \to 4\sqrt{1-y^2}\right\}\right\}$

In[122]:= **Solve[x² + 16 y² == 16, y] /. x → 0**

Out[122]= {{y → -1}, {y → 1}}

In[123]:= **volume = $\int_{-1}^{1}\int_{-4\sqrt{1-y^2}}^{4\sqrt{1-y^2}}\int_{0}^{x+4} dz\, dx\, dy$**

Out[123]= $16\,\pi$